

Patrick Norton

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Education

University of Toronto

PhD, Electrical & Computer Engineering

Toronto, ON

Sep 2026–

Reed College

BA, Mathematics & Computer Science

GPA: 3.78

Portland, OR

Aug 2021–May 2025

Relevant Courses Taken

- Computability & Complexity
- Algorithms
- Abstract Algebra
- Lie Algebras
- Deep Learning
- Linear Algebra (proof-based)
- Number Theory
- Computer Systems
- Topology
- Vector Calculus

Publications

- [1] Mark Galassi and **Patrick Norton**. *femtotime*. Computer Software. USDOE National Nuclear Security Administration (NNSA), May 2024. doi: 10.11578/dc.20240517.1. URL: <https://github.com/lanl/femtotime>.

Honours & Awards

Phi Beta Kappa Member: Phi Beta Kappa

2025

Commendation for Excellence in Academic Performance: Reed College

2021–25

Florence H. Leslie Math Prize: Greenhills School

2021

Experience

Academic

Vector Institute and the University of Toronto

Internship (mentored by Nicolas Papernot & Olive Franzese-McLaughlin)

Toronto, ON

Sep 2025–Sep 2026

- Research on validating fairness of machine-learning models.
- Used zero-knowledge proofs to be able to validate fairness without leaking public information.
- **Learning outcomes:** Computer-science research skills, mathematics of machine learning, applying zero-knowledge proofs.

Reed College

Senior Thesis

Portland, OR

Aug 2024–May 2025

- Expository thesis on probabilistically-checkable proofs and zero-knowledge proofs.
- Presented results from multiple recent papers in cryptography and complexity theory.
- **Learning outcomes:** Presenting mathematics in an accessible way, probabilistically-checkable proofs, zero-knowledge proofs, algebrization.

Reed College*Grader, Math 332***Portland, OR***Jan 2024–May 2024, Jan 2025–June 2025*

- Graded Math 332, Reed's abstract algebra course.
- **Learning outcomes:** Reading mathematical writing, abstract algebra skills.

Reed College*Peer tutor, Math***Portland, OR***August 2024–December 2024*

- Ran peer tutoring for students in Reed's first and second-year math courses.
- **Learning outcomes:** Tutoring, explaining mathematics, working with students.

Reed College*Internship***Portland, OR***May 2024–August 2024*

- Worked with Prof. Zaji Daugherty on decomposing Lie algebras into highest-weight spaces.
- Generated decompositions for the symmetric algebra.
- Presented research results at a poster session.
- **Learning outcomes:** Knowledge of Lie algebras, learning about the process of math research, combinatorial algebra skills.

Reed College*Grader, CS 221***Portland, OR***August 2022–December 2023*

- Graded CS 221, Reed's second-year computer science course, focusing on C++ and assembly.
- **Learning outcomes:** Debugging code, learning floating-point internals.

Los Alamos National Laboratory*Internship***Los Alamos, NM***May 2022–Aug 2022, May 2023–Aug 2023*

- Worked on DIORAMA, a nuclear-and-satellite simulation program used by the US government.
- Helped modernize code and remove technical debt, as well as refactoring the testing framework.
- Created a library for femtosecond-precision timekeeping (`femtoime [1]`), able to work with both UTC and GPS time as well as converting between the two.
- **Learning outcomes:** Debugging and modernizing production C++, familiarization with debugging systems, learning about UTC and GPS time, learning about different location schemes.

Dartmouth University*Internship***Hanover, NH (remote)***May 2021–Aug 2021*

- Assisted Prof. Sean Smith's graduate students with creating a debugger for a parser.
- **Learning outcomes:** Introduction to C, knowledge of parser systems.

Greenhills School*Senior Project***Ann Arbor, MI***Jan 2021–Apr 2021*

- Implemented a lock-free concurrent vector in Rust.
- **Learning outcomes:** Introduction to concurrent programming and algorithms, familiarization with concurrent debugging techniques.

Miscellaneous.....**Math Corps at U(M)***Community Assistant***Ann Arbor, MI***July 2025–Aug 2025*

- Assisted teaching mathematics to middle and high schoolers.
- Helped grade homework and come up with problem sets.
- Helped foster a positive community among campers.

Blue Lake International*Musician***Blue Lake, MI***July 2018 & July 2019*

- Played trombone in an international concert band.
- Toured in Germany, France, the Netherlands, and Poland.

Relevant Skills & Languages

- Programming Languages

- Fluent: Rust, Java, Lisp, C++
- Proficient: C, Python, Haskell, OCaml, MIPS assembly
- Linux, MacOS, Emacs, L^AT_EX